THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 60 km/h	90/1.5.
2	**(a)** 50 km/h	100/2.
3	**(d)** speed	Rise over run.
4	**(c)** 9 km/h	1.5 x 6.
5	**(b)** 2 h	60/30.
6	**(a)** 5 m/s	20/4.
7	**(c)** 10 m/s	36 x 5/18.
8	**(c)** 8 m/s	100/12.5.
9	**(d)** faster	10 m/s versus 5 m/s.
10	**(a)** 90 km/h	25 x 18/5.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	150/2.5 = 60 km/h (1); 60 x 5/18 = 16.7 m/s (approx.) (1).	1+1
12	Effort and gravity act against uphill (1); Speed changes so motion is non-uniform (1).	1+1
13	Larger angles can change motion (1); Time period measurement is easier (1).	1+1
14	Time = 225/90 (1); 2.5 h (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	40 min = 2/3 h (1); 12 / (2/3) = 18 km/h (1); 5 m/s (1).	1+1+1
16	Going 3/5 = 0.6 h (1); Returning 3/3 = 1 h (1); Total 1.6 h (1).	1+1+1
17	First 0.5 m/s (1); Second 0.75 m/s (1); Speed increased (1).	1+1+1
	OR: 200/20 = 10 m/s (1); 10 x 18/5 (1); 36 km/h (1).	1+1+1
18	Traffic (1); Signals (1); Stops and starts (1).	1+1+1
	OR: Accuracy (1); Avoid human delay (1); Fair results (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total time 4 h (1); Average 240/4 = 60 km/h (1); Running time 3.5 h (1); 240/3.5 = 68.6 km/h (1); Running speed is higher (1).	1+1+1+1+1
	OR: Friction and air resistance (1); Amplitude decreases (1); Spring or weights give energy (1); Regular winding (1); Clock continues (1).	1+1+1+1+1
20	40 km in first 2 h (1); Horizontal for 1 h (1); 60 km more in 2 h (1); Total 100 km in 5 h (1); Average 20 km/h (1).	1+1+1+1+1
	OR: Cheetah 25 m/s (1); Car 20 m/s (1); Cheetah faster (1); Convert 25 m/s to 90 km/h (1); Compare (1).	1+1+1+1+1

Mistakes to avoid:

- Dividing time by the number of oscillations the wrong way; time period = **total time ÷ number of oscillations**.
- Writing that a heavier bob has a larger time period; the period depends on **length**, not on mass.
- Forgetting to convert units before calculating (minutes to hours, km/h to m/s).
- Using $\times \frac{18}{5}$ instead of $\times \frac{5}{18}$ when converting km/h to m/s.
- Calculating average speed as the average of two speeds instead of *total distance divided by total time*.
- Reading a horizontal line on a distance-time graph as 'moving fast'; it means the object is **at rest**.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** half	Time inversely proportional.
2	**(a)** the object at rest 30 m away	Distance fixed.
3	**(c)** 200 m	20 m/s x 10 s.
4	**(b)** double	Proportional.
5	**(a)** speed increasing	Slope increases.
6	**(b)** 2.5 s	60/24.
7	**(b)** 15	54 x 5/18.
8	**(c)** distance rising and then falling	Displacement returns.
9	**(c)** 40 oscillations	60/1.5.
10	**(d)** 10 m/s	100/10.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Measure the distance and time to cover it (1); Speed = distance/time (1).	1+1
12	Multiply by 5/18 (1); 1000 m per 3600 s (1).	1+1
13	T = 50/25 = 2 s (1); 180/2 = 90 oscillations (1).	1+1
14	Speed keeps changing (1); Total distance divided by total time (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Time 2 h + 1 h = 3 h (1); Distance 120 km (1); 40 km/h (1).	1+1+1
16	Container with a small hole (1); Mark water level (1); Time between marks (1).	1+1+1
17	Speedometer shows instantaneous speed (1); Average is total distance/time (1); They may differ (1).	1+1+1
	OR: A: 50 km/h (1); B: 50 km/h (1); Equal speeds (1).	1+1+1
18	72 km/h = 20 m/s (1); 36 km/h = 10 m/s (1); 10 < 15 < 20 (1).	1+1+1
	OR: Measure a 100 m track (1); Time your walk (1); Divide (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	As L increases, T increases (1); Longer pendulum is slower (1); Mass kept same (1); Time 20 oscillations for accuracy (1); Conclude T depends on length only (1).	1+1+1+1+1
	OR: Start gun triggers (1); Sensors at the finish (1); Time to 0.01 s (1); Human reaction slower (1); Accuracy (1).	1+1+1+1+1
20	Sundial: gnomon and shadow (1); Water clock: flow of water (1); Sundial works by day only (1); Water clock works day and night (1); Both less accurate (1).	1+1+1+1+1
	OR: Two bobs of different masses (1); Same length (1); Count 20 oscillations each (1); Same time (1); Mass has no effect (1).	1+1+1+1+1

Mistakes to avoid:

- Dividing time by the number of oscillations the wrong way; time period = **total time ÷ number of oscillations**.
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- Forgetting to convert units before calculating (minutes to hours, km/h to m/s).
- Using $\times \frac{18}{5}$ instead of $\times \frac{5}{18}$ when converting km/h to m/s.
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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** non-uniform motion	Varying speed.
2	**(b)** the second hand of a clock	Constant speed.
3	**(c)** 1.5 h	0.5 + 1.
4	**(c)** measuring 20 oscillations	Reduces error.
5	**(a)** 50 m	5 m/s x 10.
6	**(d)** 90 km/h	90 km/h = 25 m/s.
7	**(a)** increases	Longer swings slower.
8	**(a)** 2 km	4 x 0.5.
9	**(a)** 3 m/s	18 km/h = 5 m/s; 3 m/s smallest.
10	**(b)** remains same	Independent of mass.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total distance 20 m (1); 20/10 = 2 m/s (1).	1+1
12	8 min = 2/15 h (1); 2 / (2/15) = 15 km/h (1).	1+1
13	Subtract the initial reading from the final (1); Difference is distance (1).	1+1
14	Quartz oscillates regularly (1); Sundial depends on sunlight (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Time 20 oscillations (1); T = 36/20 (1); 1.8 s (1).	1+1+1
16	16/10 = 1.6 s (1); 32/20 = 1.6 s (1); T = 1.6 s in both cases (1).	1+1+1
17	Equal swing times (1); Count swings (1); Clock use (1).	1+1+1
	OR: 8/2 = 4 m/s (1); 16/4 = 4 m/s (1); Uniform (1).	1+1+1
18	Repeats in equal times (1); Speed changes (1); Fast at the lowest point (1).	1+1+1
	OR: Needs clear shadow (1); No sunlight (1); Shadow unclear (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Start: speed increasing (1); Straight track: uniform (1); Stop: speed decreasing (1); Whole journey non-uniform (1); Graph is curved at ends (1).	1+1+1+1+1
	OR: Time out 2 h (1); Time back 1 h (1); Total distance 120 km (1); 120/3 = 40 km/h (1); Not 45 (1).	1+1+1+1+1
20	Walk 900/600 = 1.5 m/s (1); Run 900/180 = 5 m/s (1); Total 1800 m in 780 s (1); 2.3 m/s (1); Between the two (1).	1+1+1+1+1
	OR: Plot points (1); Join (1); Slope gives speed (1); Flat part is rest (1); Compare sections (1).	1+1+1+1+1

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